

2010. Post-2005, there is a 65% and 84% correlation between Baa and equities and Caa and equities, respectively. Thus, *corporate bonds are (scaled-down) versions of equity returns*.

Some of the same intuition, then, that explains the equity premium can be extended to explain the credit spread puzzle.²⁸ Slow growth or recessions are bad times for both equities and bonds. (This is the same macro channel for equities that we covered in chapter 8.) But these same bad times for equities are also bad times for corporate bonds. There are more defaults during these bad times because of the poor performance of firms during recessions, and equity holders choose to liquidate their firms earlier to preserve firm value. In the context of the structural credit models of section 4.2, the boundary of liquidation *rises* during recessions. Bondholders also recover less when firms default during recessions than during booms. Thus, corporate bond investors are exposed to the same types of macro bad times risks as equity holders. The high credit spreads compared to realized defaults are due to investor risk aversion, since the losses on corporate bonds come right during recessions—precisely when investors can afford them least. To harvest the corporate bond risk premium, as modest as it is, investors must be able to stomach strong losses during economic slowdowns.

4.4. LIQUIDITY RISK IN CORPORATE BONDS

Liquidity risk—the ability to easily trade at low cost—is an important factor for corporate bond returns. Collin-Dufresne, Goldstein, and Martin (2001) argue that

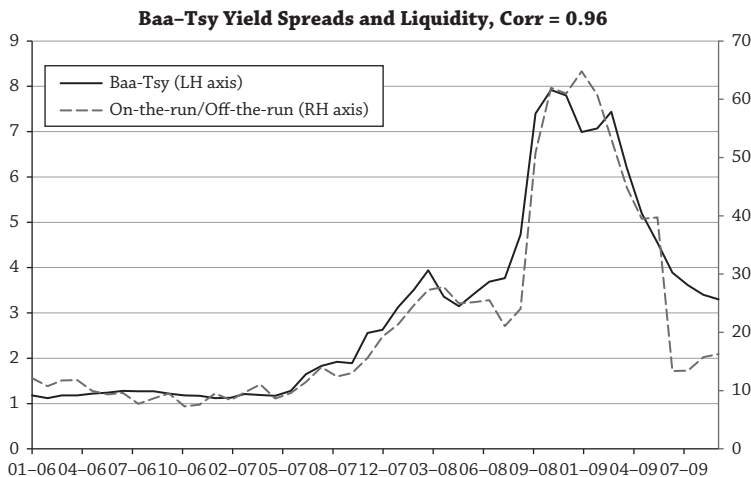


Figure 9.17

²⁸ See Chen, Collin-Dufresne, and Goldstein (2007) and Chen (2010) for models along these lines.